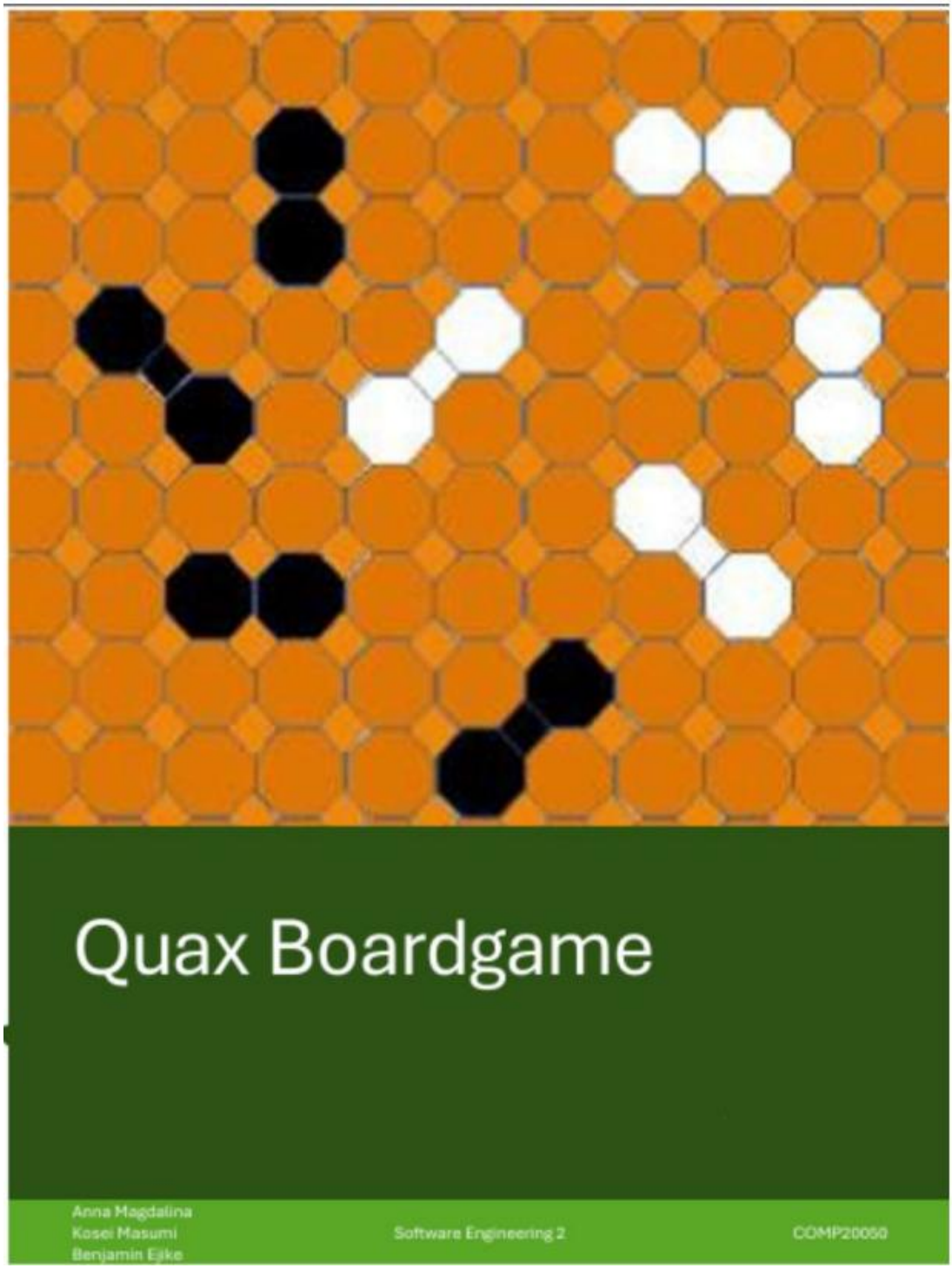

Quax Boardgame
Anna Magdalina
Kosei Masumi
Benjamin Ejike
Software Engineering 2
COMP20050

1 Launch Instructions

1.1 System Requirements

JDK Version	Java Development Kit (JDK) 24
JavaFX Version	JavaFX SDK 25
JAR File	quax.jar
OS	Any system capable of running Java applications

1.2 Running the Game

Open a terminal or command prompt in the directory containing the JAR file and run the following command:

```
java --module-path "javafx-sdk-25/lib"
     --add-modules javafx.controls,javafx.graphics
     -jar quax.jar
```

This command launches the JavaFX application and opens the Quax game window. If the JavaFX library is in a different folder, adjust the module path accordingly.

> **Note** The game is provided as a runnable JAR file together with the project source code. Ensure JavaFX SDK 25 is extracted locally and the path is set correctly.

1.3 Troubleshooting

If the game does not launch correctly, check the following:

- Ensure that JavaFX SDK 25 is correctly installed.
- Ensure that the JavaFX module path in the command matches the actual folder location.
- Ensure that JDK 24 is installed and set as the active Java version.

2 Game Overview

Quax is a strategy board game played on an 11×11 grid of octagonal tiles. In this version of the game, the human player competes against a bot player.

Black (Bot)	Aims to connect the TOP side of the board to the BOTTOM side.
White (Player)	Aims to connect the LEFT side of the board to the RIGHT side.
Winner	The first player to create a continuous connection between their designated sides wins the game.

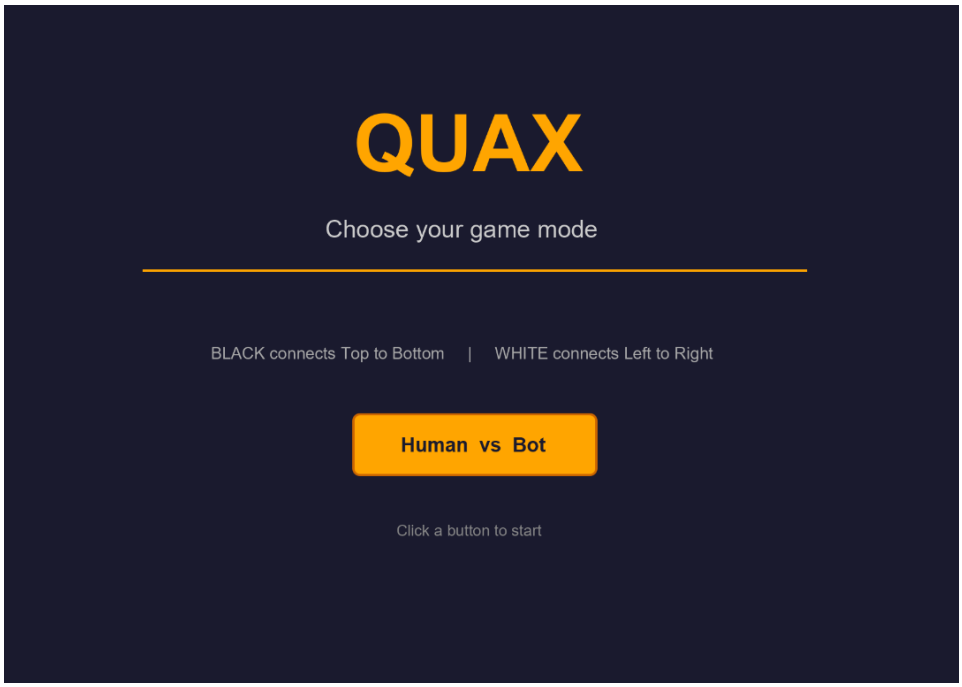
2.1 Game Interface

When the game launches, the main window displays:

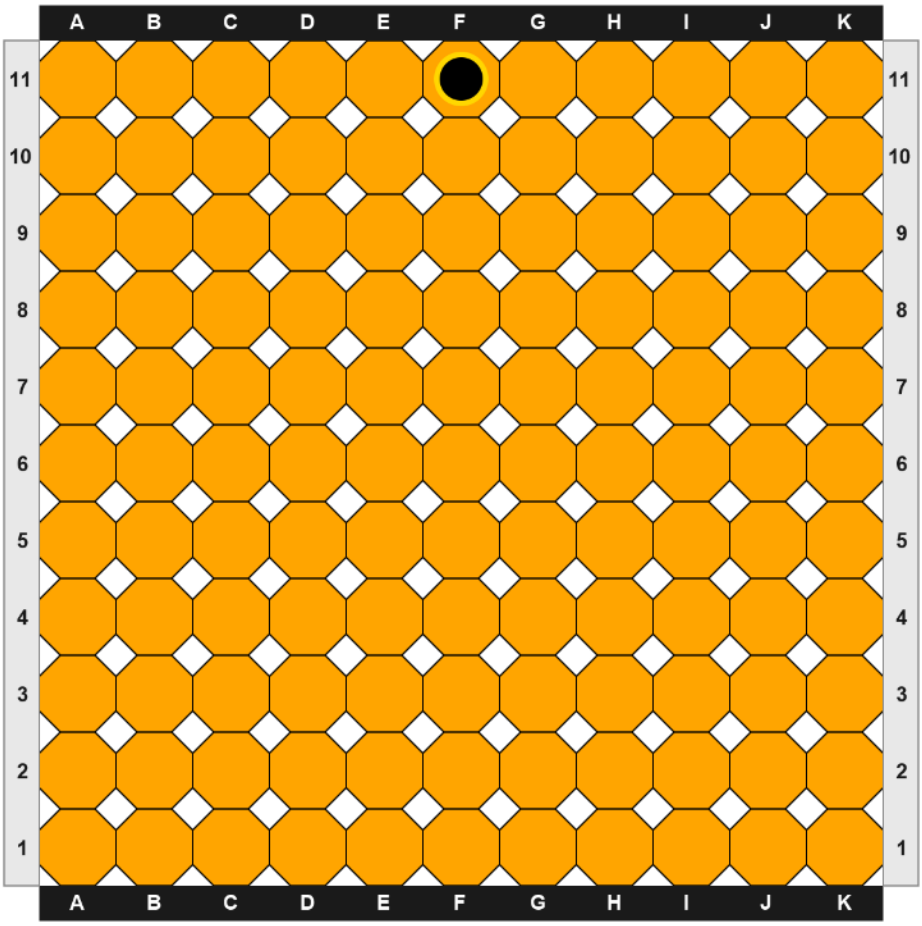
- A small opening page with a button to start the game.
- The 11×11 Quax board with octagonal tiles and rhombic positions.
- A turn indicator showing whose turn it is (human player or bot).
- The user interacts with the game by clicking on tiles using the mouse.

3 How to Play

3.1 Starting the Game



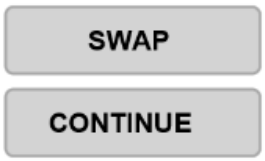
- The game starts with a start screen showing short rules about win detection. Click the Human vs Bot button to begin.



- The game begins with Black (the bot) making the first move automatically.

3.2 The Pie Rule

- After the bot's first move, White (you, the player) is given the option to use the **Pie Rule**.

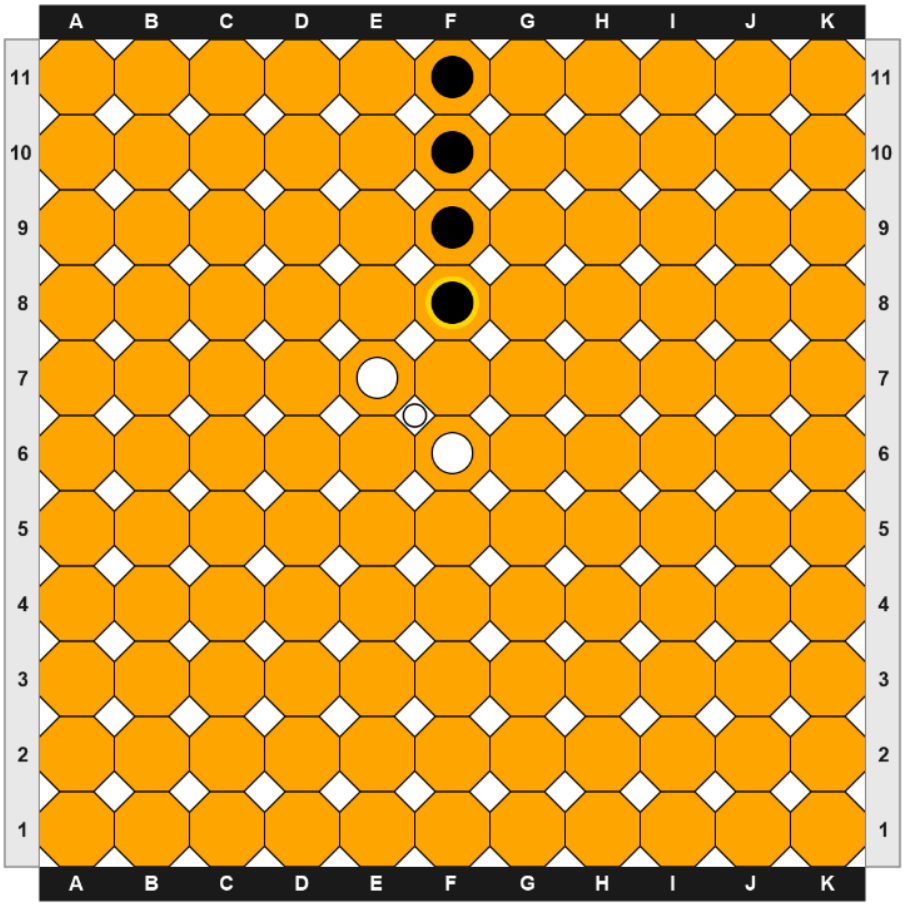


SWAP — All stones on the board swap colours. The bot will make the next move as White, and you can place your tile afterwards as Black.

CONTINUE — Decline the swap. You remain as White and may place your tile immediately.

> **Note** The Pie Rule buttons disappear after one click and never reappear.

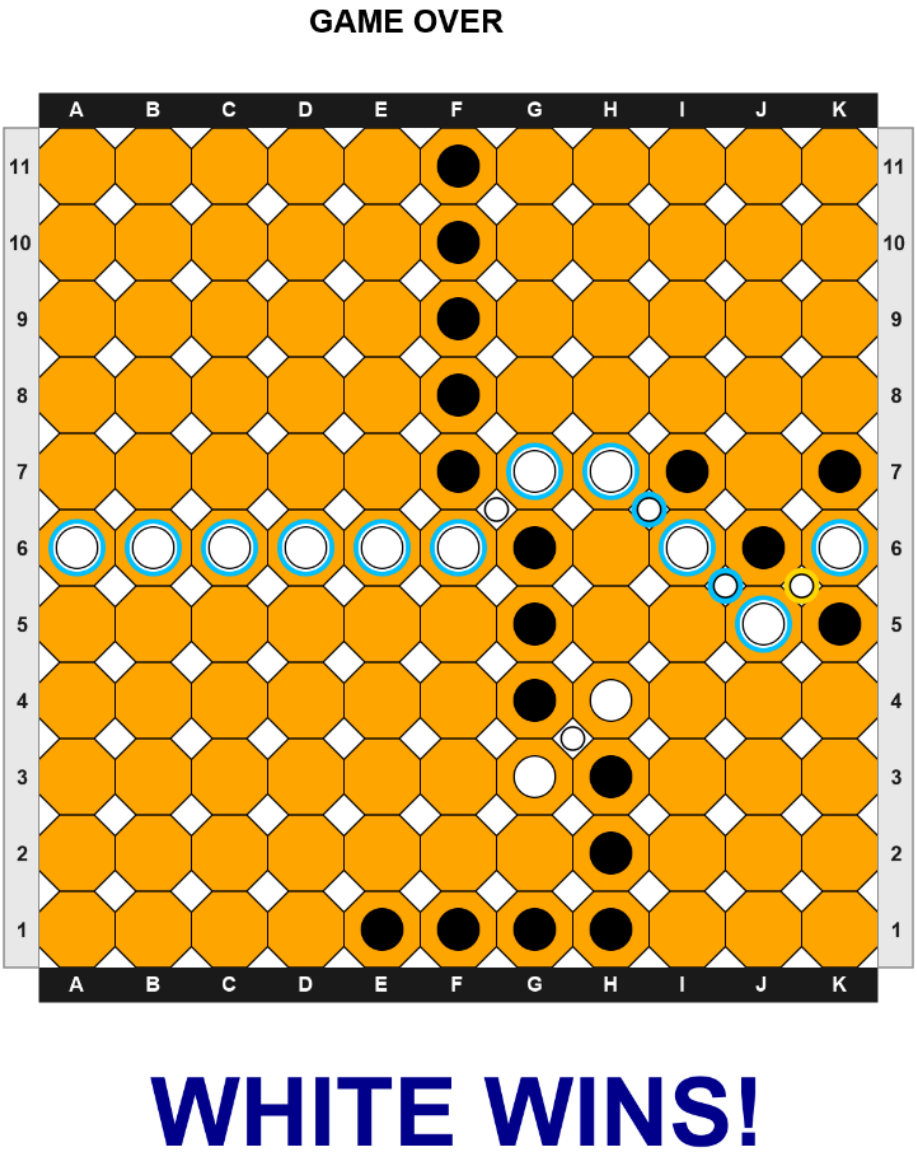
3.3 Gameplay



- To place a **stone**: click any empty orange octagonal cell during your turn.
- To place a **rhombic tile**: click a diamond-shaped gap between two of your diagonally adjacent stones. Both corner cells must contain your colour.
- You and the bot will continue to take turns until a winning chain is detected.

> **Note** You cannot place tiles during the bot's turn. All input is locked until the turn indicator shows your colour.

3.4 Winning



- Once someone wins, the game displays a highlighted winning chain and shows the winning colour.
- No more tiles can be placed after the game ends.

Good luck and enjoy the game!